ACME ASIA 1,024-GPU HGX B300 AI Factory

Phased Solution Architecture Proposal

64 -> 128 -> 1,024 GPUs | 8 -> 16 -> 128 HGX B300 systems

Document	**Solution Architecture Proposal - customer review baseline**
Revision	1.1
Prepared for	ACME ASIA
Prepared by	ViableCloud UK
Scope	AI infrastructure and enterprise solution architecture
Issue date	1 September 2026
Issue status	Architecture complete; explicit vendor/customer confirmations remain

Detailed data-centre facilities engineering is outside the scope of this Solution Architecture and is handed off to qualified facilities engineers in coordination with Supermicro, NVIDIA NVIS and the selected data-centre operator.

1. Executive Summary

This proposal defines the target solution architecture for a phased NVIDIA HGX B300 AI Factory and closes the principal architectural gaps identified in the original source material spreadsheet: GPU east-west fabric topology, converged north-south and storage networking, phased WEKA capacity, security and firewall boundaries, out-of-band management, platform control, and enterprise operations.

The target state is 128 HGX B300 systems (1,024 B300 GPUs), delivered as three production-valid population states of 8, 16 and 128 systems. The solution uses Quantum-X800 InfiniBand for the dedicated GPU east-west fabric, Spectrum-X Ethernet for converged north-south/storage/platform traffic, WEKA for shared AI storage, dedicated OOB management, an HA security boundary, and Rafay as the prominent cloud-hosted Kubernetes control plane alongside Slurm for tightly coupled training and HPC workloads. The architecture is sufficiently closed for customer review and downstream engineering, subject to the explicit Supermicro, WEKA and ACME confirmations listed in Section 16.

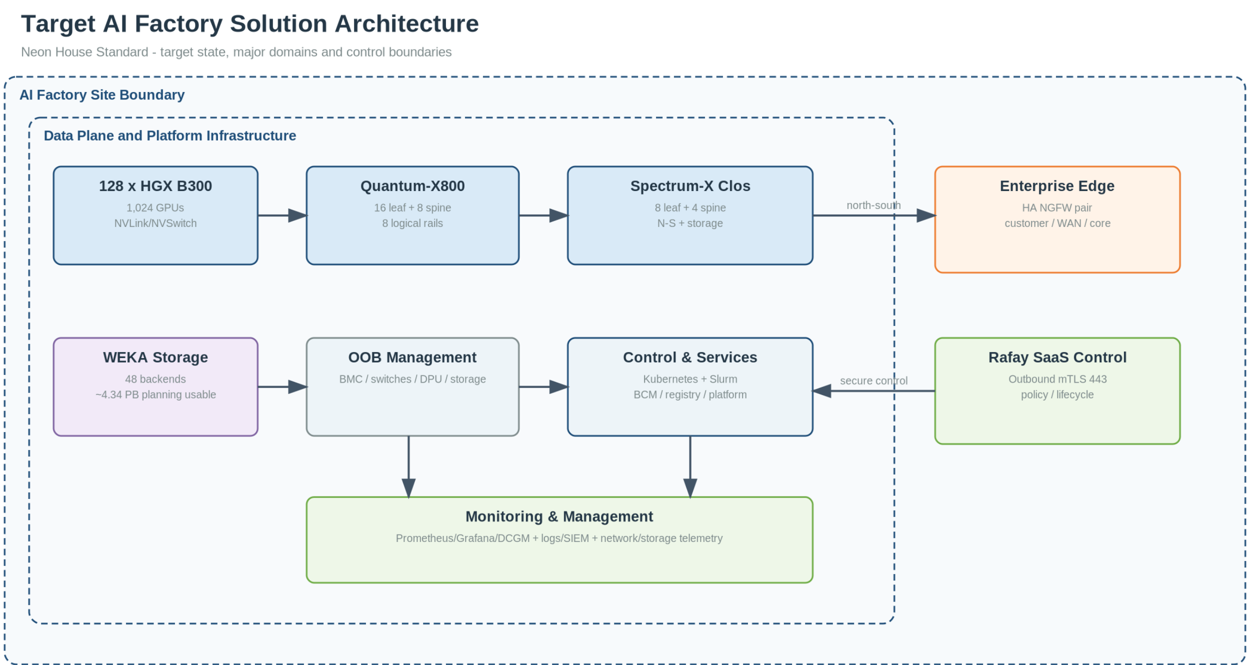

Figure 1 - Target solution architecture

2. Scope and Architecture Principles

- One target architecture, three population states: Phase 1 and Phase 2 are not disposable PoCs.
- Customer requirements are separated from source-design assumptions, SME decisions, vendor-validation items and unresolved customer inputs.
- Architecture decisions, derived quantities and detailed physical-network evidence are traceable to governed requirements, equipment knowledge and deterministic engineering outputs.
- High-speed storage is physically converged onto Spectrum-X but remains logically isolated by VRF/VLAN, policy and QoS.
- Detailed facilities engineering is out of scope; the proposal supplies equipment and interface requirements for the downstream facilities workstream.
- Procurement SKUs and optics/cable media are frozen only where sufficiently grounded; rack-distance-dependent media remain a physical-design dependency.

3. Architecture Design Basis - Key Decisions

Domain	Decision	Status
Compute	HGX B300 NVL8, 8 GPUs/node; 8 -> 16 -> 128 systems	DECIDED
GPU E-W	Quantum-X800 XDR InfiniBand; 8 logical rails; two-tier non-blocking fabric	DECIDED
E-W switching	Final 16 Q3400 leaf + 8 Q3400 spine = 24 Q3400	DECIDED
N-S/storage	Spectrum-X Ethernet, SN5610 leaf/spine Clos, physically converged	DECIDED
N-S switching	Final 8 SN5610 leaf + 4 SN5610 spine = 12 SN5610	DECIDED
Host N-S	1 x BlueField-3 B3240/node, dual 400GbE target	VALIDATE Supermicro
Storage	WEKA NeuralMesh; ~1.09 -> 2.17 -> 4.34 PB usable planning path	DECIDED / VERIFY WEKA
WEKA population	12 -> 24 -> 48 backends, 10 x 15.36 TB NVMe planning geometry	DECIDED / VERIFY WEKA
Security	HA NGFW pair, explicit trust zones, controlled admin path, centralized logging	DECIDED
OOB	Dedicated management fabric, sized with endpoint headroom rather than 100% port use	DECIDED
Platform	Rafay SaaS controller prominent; Kubernetes + Slurm coexist	DECIDED
Bare-metal lifecycle	BCM retained for provisioning/infrastructure lifecycle; Rafay manages Kubernetes/platform lifecycle	DECIDED
Facilities	Detailed electrical, cooling, CDU, hydraulic and rack engineering	OUT OF SCOPE
External bandwidth	Customer/core/WAN bandwidth and final firewall throughput	CUSTOMER CONFIRMATION

4. Compute Architecture

- Each HGX B300 system contains eight B300 GPUs in a node-local NVLink/NVSwitch scale-up domain.
- The target network envelope is eight ConnectX-8 800G XDR interfaces for the dedicated GPU east-west fabric and one BlueField-3 B3240 with two 400GbE interfaces for north-south, storage and platform traffic.
- Any supplier system whose DPU configuration differs from that envelope is treated as a compliance deviation requiring explicit acceptance rather than silently redefining the reference architecture.

Phase	B300 systems	GPUs	CX-8 E-W endpoints	BF3 target
1 - Foundation	8	64	64	8 x B3240
2 - Expansion	16	128	128	16 x B3240
3 - Target	128	1024	1024	128 x B3240

5. Quantum-X800 East-West Fabric

- The production reference design uses a two-tier, non-blocking, rail-optimized XDR InfiniBand topology.
- The final 128-node population is placed inside a 144-node/two-SU fabric envelope, retaining 16 unused server positions without redesigning the network.

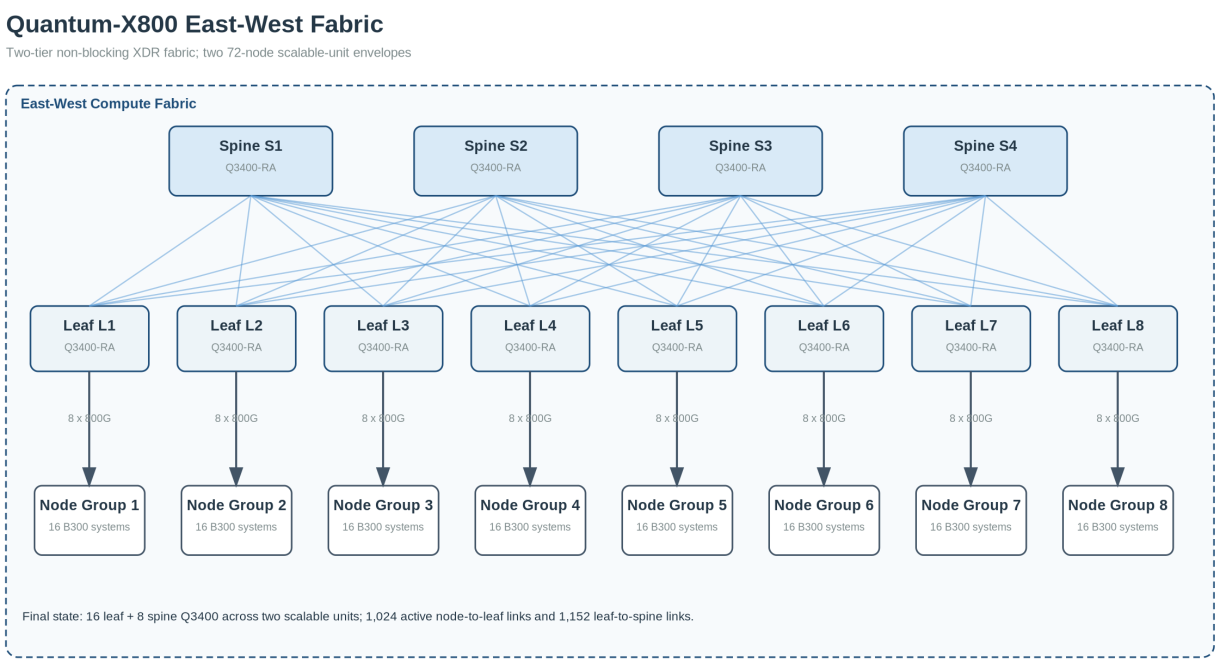

Figure 2 - Quantum-X800 two-tier topology

Metric	Final value
Q3400 leaf switches	16
Q3400 spine switches	8
Total Q3400	24
Active server-to-leaf links	1,024 x 800G XDR
Leaf-to-spine links	1,152 x 800G XDR
Final fabric node capacity	144 B300 nodes
Active population	128 B300 nodes
Leaf blocking ratio	1:1 / non-blocking

Recommended phasing is reference-compliant at the scalable-unit boundary: Phase 1 installs one complete 72-node SU fabric (8 leaf + 4 spine = 12 Q3400) even though only eight B300 nodes are initially populated; Phase 2 reuses that fabric for sixteen nodes; Phase 3 adds the second complete SU fabric for 24 Q3400 total. The final physical realization reconciles 1,024 active server-to-leaf links and 1,152 leaf-to-spine links (2,176 exact links) with device/port-level traceability in the supporting engineering evidence. This intentionally invests in the stable fabric envelope early and avoids repeated east-west topology replacement.

6. Spectrum-X Converged North-South / Storage Clos

- The final converged fabric uses eight SN5610 leaves and four SN5610 spines. Every leaf has 64 native 800G ports: 32 are allocated to ISLs and 32 form the endpoint-facing envelope.
- Each leaf connects to every spine with eight 800G ISLs, producing 32 ISLs per leaf and 64 leaf-facing ISLs per spine.
- Endpoint-facing 800G cages are broken out to 2 x 400G where required for BlueField and WEKA attachment.

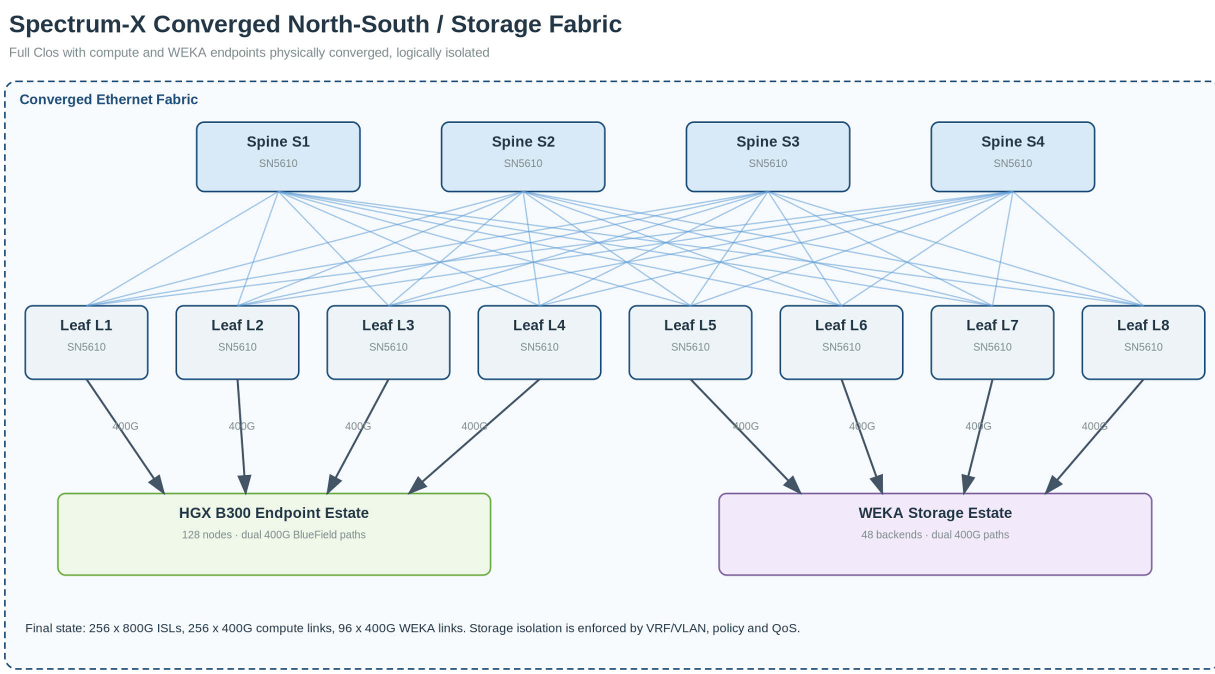

Figure 3 - Converged SN5610 Clos

Per-leaf port envelope	800G port-equivalents
Spine ISLs	32
B300 / BlueField	16
WEKA backends	6
Platform/control services	1
External/core/firewall envelope	1
Infrastructure/in-band management	1
Reserved / expansion	7
Total	64

- Across eight leaves the design allocates 128 native 800G port-equivalents to the 256 x 400G B300/BlueField links and 48 native 800G port-equivalents to the 96 x 400G WEKA links, while retaining capacity for platform services, customer/core connectivity and growth.
- The final physical realization reconciles all 608 links (256 compute, 96 WEKA and 256 leaf-spine ISLs) to exact device and port identities. Exact external/firewall ports remain dependent on the ACME edge-bandwidth requirement rather than being populated solely because switch radix is available.

7. WEKA Storage Architecture

- The source 4 PB / 1 PB inconsistency is resolved through one scale-out namespace and a phased storage population.
- The governed 12-backend configuration is retained as the customer-facing sizing anchor.
- Using 10 x 15.36 TB NVMe devices per backend under the established reserve/protection assumptions it yields approximately 1.086 PB usable.
- The same bounded geometry is projected to 24 and 48 backends for the later phases.

Phase	WEKA backends	Raw flash (TB)	Planning usable (TB)	400G links	SN5610 800G equivalents
1	12	1,843.2	~1,086	24	12
2	24	3,686.4	~2,172	48	24
3	48	7,372.8	~4,345	96	48

- The 250 GB/s read and 100 GB/s write figures are interpreted as aggregate final-cluster targets pending confirmation of benchmark semantics.
- They are not interpreted per B300 node.
- The 24- and 48-backend states remain bounded derived candidates until WEKA verifies the exact SSD SKU, protection geometry, usable capacity, performance/IOPS and licensing metric
- This qualification does not prevent the design from driving the network, phasing and architecture-level BoM.

8. Security and External Connectivity

Security is a first-class architecture domain. The design inserts a redundant HA next-generation firewall pair between the ACME core/WAN/customer boundary and the AI Factory service edge. The trust-zone, privileged-access and logging architecture is fixed; the exact firewall vendor/model and throughput class remain dependent on ACME external-bandwidth and security-service requirements.

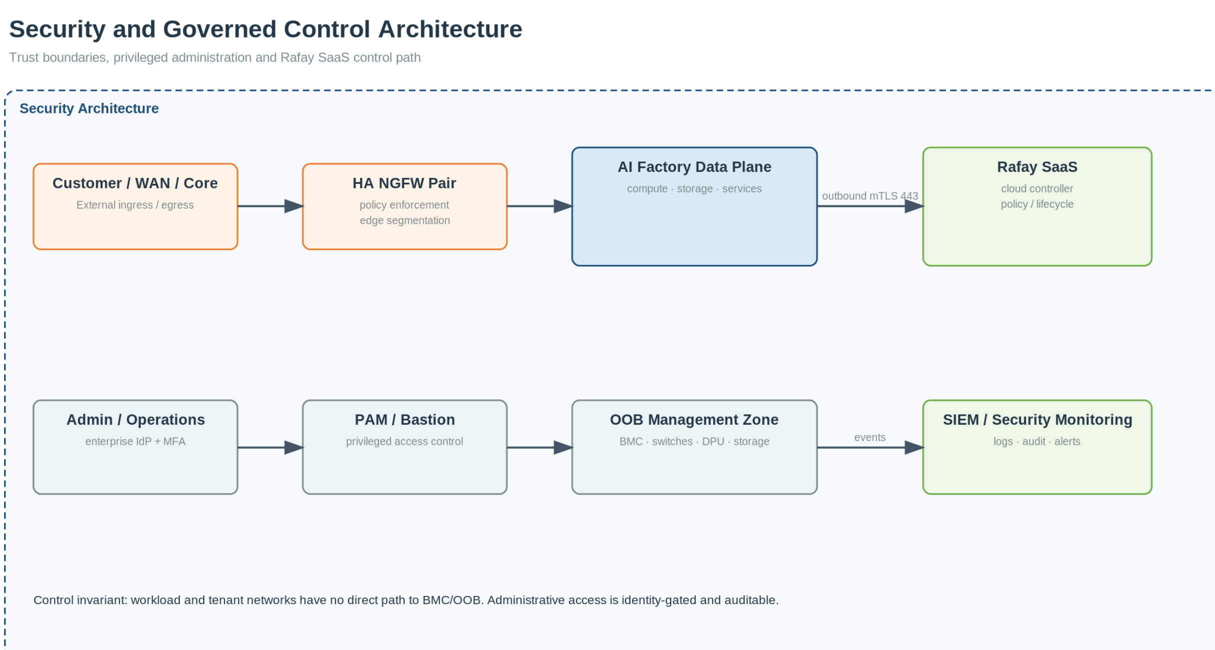

Figure 4 - Trust-zone and Rafay control-path architecture

Zone	Purpose	Key policy
EDGE / CUSTOMER	Customer, WAN, Internet and upstream DC connectivity	Only explicitly published services; firewall-enforced
PLATFORM / CONTROL	Rafay agents, Kubernetes, Slurm, provisioning, registry	Restricted administrative/service access
COMPUTE	B300 workload interfaces	No direct BMC/OOB reachability
STORAGE	WEKA data path	Only approved compute/platform storage flows
OOB	BMC, switch, DPU, storage and infra management	PAM/bastion/admin-only access
OBSERVABILITY / SIEM	Metrics, logs, security events	Receive telemetry; tightly controlled administration

For Rafay SaaS, the managed Kubernetes environment initiates the control channel outbound over TCP 443 using mutual TLS. No inbound firewall opening is required for the Rafay management control channel. Customer policy may additionally restrict destinations by proxy/FQDN/IP allowlisting.

9. Out-of-Band Management Architecture

- The source OOB design consumed its calculated access capacity without useful headroom.
- The revised architecture separates compute OOB from infrastructure OOB and sizes the access layer with reserve capacity, while retaining redundant aggregation.
- The final planning baseline is 21 SN2201-class access switches plus two 100G-capable aggregation switches; the exact aggregation model remains a procurement decision.

Final OOB element	Planning quantity	Basis
SN2201 compute OOB access	18	768 compute management endpoints / 864 ports = ~88.9% utilization
SN2201 infrastructure OOB access	3	Q3400, SN5610, WEKA, control/observability and security management endpoints with reserve
Total SN2201 access	21	Dedicated access layer
OOB aggregation	2	Redundant 100G-capable aggregation switches; exact model procurement decision
Access uplinks	42 x 100G physical links	Each SN2201 dual-homed to aggregation

- The OOB network is not reachable from tenant/workload networks.
- Administrative entry is via enterprise identity, MFA and PAM/bastion controls.
- BMC, DPU management, switch management and storage management remain distinct from the high-speed data fabrics.

10. Enterprise Software and Control Architecture

- Rafay is positioned as the prominent enterprise Kubernetes control and policy plane, while BCM provides bare-metal/infrastructure lifecycle services and Slurm remains available for tightly coupled distributed training and HPC-style batch workloads.
- Kubernetes and Slurm therefore coexist rather than forcing all workloads into one scheduler.

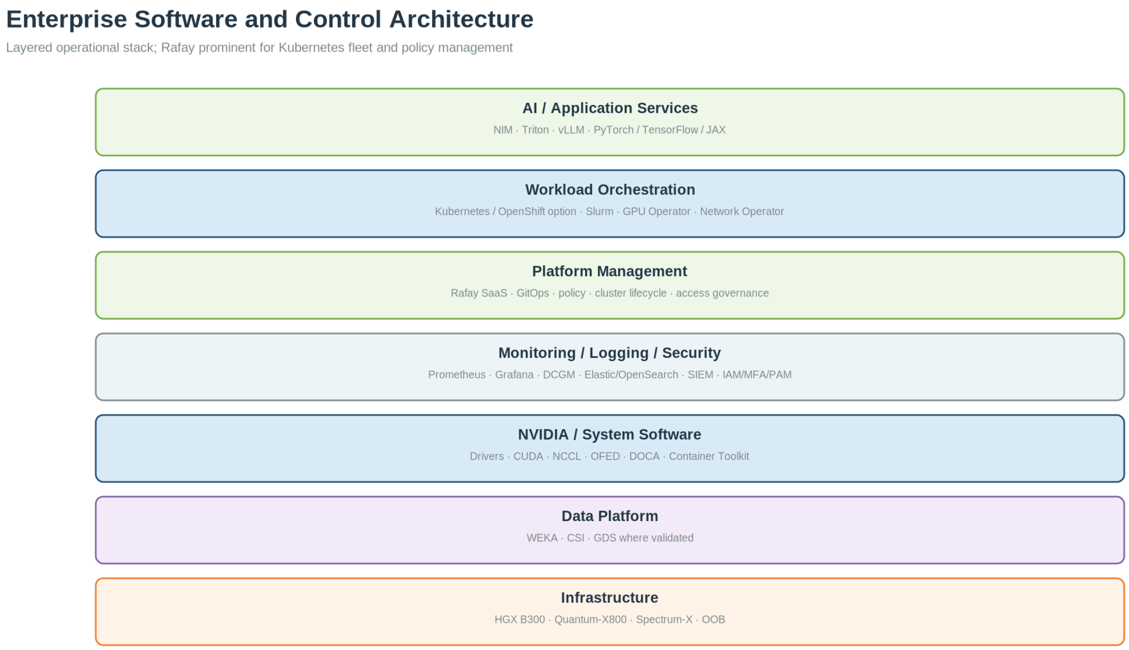

Figure 5 - Enterprise software stack

Layer	Baseline
Platform control	Rafay SaaS Controller; Rafay operator/agent in managed Kubernetes clusters
Bare-metal lifecycle	NVIDIA Base Command Manager (BCM) / equivalent provisioning integration
Container orchestration	Upstream enterprise Kubernetes baseline; OpenShift remains an optional customer substitution
HPC/batch scheduling	Slurm HA controllers + accounting integration
GPU/network operators	NVIDIA GPU Operator; Network Operator where applicable
NVIDIA system software	Drivers, CUDA, NCCL, OFED/network software, DOCA where BlueField functions require it, Container Toolkit
Storage software	WEKA NeuralMesh, CSI integration, snapshots/encryption/QoS according to licensed features
Observability	Prometheus, Grafana, DCGM/DCGM Exporter, alerting
Logging/security	Elastic/OpenSearch-class centralized logging + SIEM integration

Platform services	DNS, NTP, registry, Git source-of-truth, secrets, PKI, backup/config repository

- The control-plane hardware planning baseline is ten x86 service nodes: 2 BCM/provisioning, 3 Kubernetes administrative/control nodes, 3 Kubernetes user/platform-service nodes and 2 Slurm control nodes.
- A separate six-node observability/logging service pool is the final-scale planning baseline.
- Retention-driven observability sizing can be refined without consuming B300 compute capacity.

11. Monitoring and Management System Architecture

- The AI Factory requires a dedicated Monitoring and Management System that provides operational visibility across hardware, fabrics, storage, platform services, workloads and security controls.
- The architecture separates monitoring from control authority: telemetry is collected centrally for health, performance, capacity, audit and incident response, while privileged infrastructure administration remains governed through the OOB and PAM path.
- Monitoring coverage includes HGX B300 node and GPU health.
- NVIDIA DCGM telemetry; operating-system metrics
- Quantum-X800 and Spectrum-X switch, port, error and congestion telemetry
- WEKA capacity, throughput, latency, protection and node health; Kubernetes control-plane, node and workload telemetry Slurm scheduler and job health
- Rafay platform events; and firewall, identity and privileged-access audit events.
- Note: NetQ, NMX, UFM, Netbox, Netris integration and other architecture for DCIMS integration will be offered in a follow up architecture document

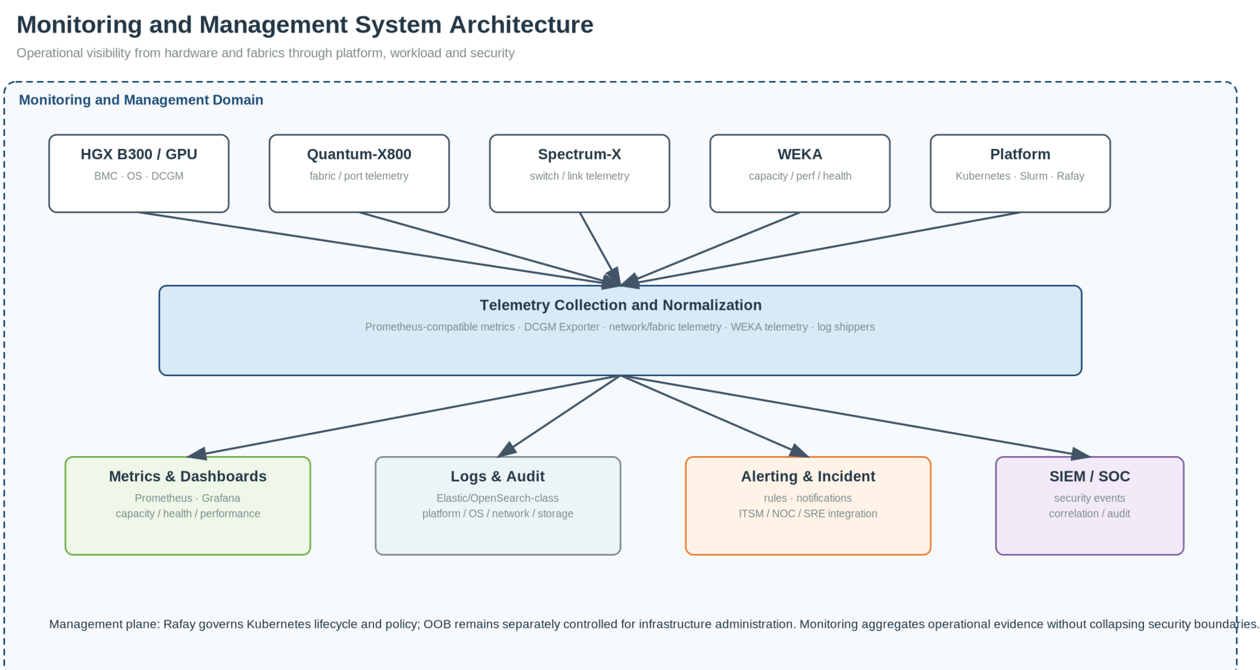

Figure 6 - Monitoring and Management System architecture

- The monitoring plane is organized around four outputs: metrics and dashboards for capacity/health/performance, centralized logs and audit, alerting/incident workflows for NOC/SRE/ITSM integration, and security-event forwarding to the enterprise SIEM/SOC.

- Prometheus-compatible collection and Grafana-style visualization form the indicative metrics baseline, with DCGM for GPU telemetry and Elastic/OpenSearch-class services for centralized logs.
- Final retention periods, enterprise ITSM integration and SIEM destination remain customer confirmation items.

12. Phased Deployment and Evolution

Phasing is a core design feature and is preserved explicitly. Every phase is production-valid and evolves by adding capacity rather than replacing the fundamental architecture.

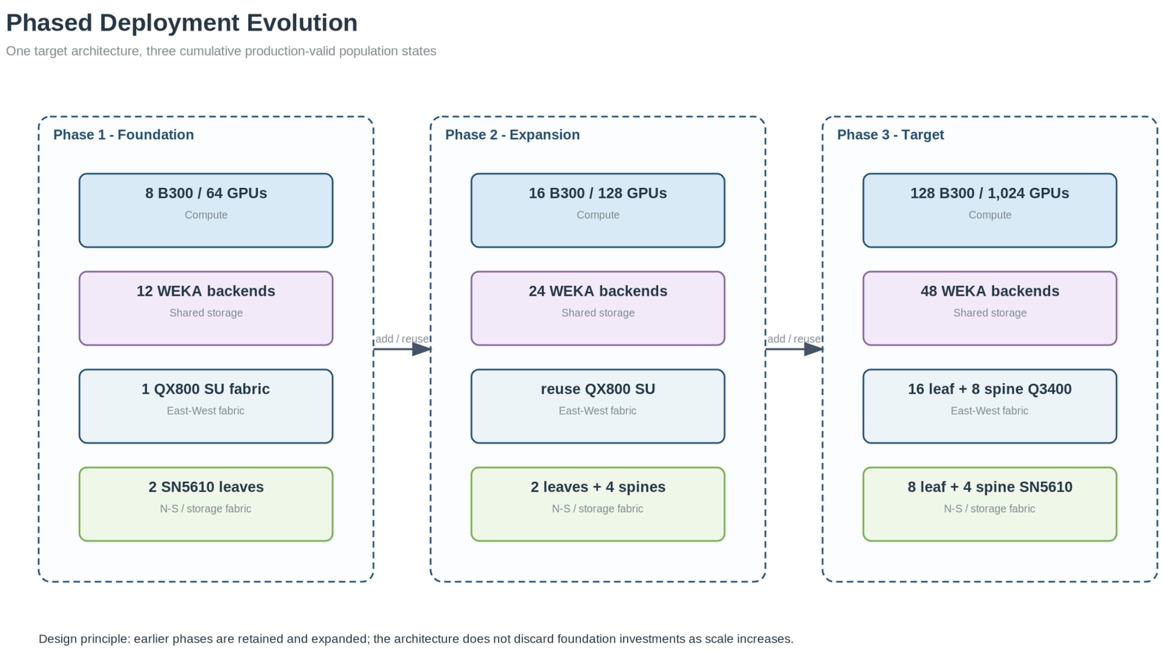

Figure 7 - Phased architecture evolution

Domain	Phase 1 - 8 B300	Phase 2 - 16 B300	Phase 3 - 128 B300
Compute	8 / 64 GPUs	16 / 128 GPUs	128 / 1,024 GPUs
Quantum-X800	12 Q3400 (1 full SU fabric)	12 Q3400 reused	24 Q3400 (2 SU fabric)
Spectrum-X	2 SN5610 leaves	2 leaves + 4 spines	8 leaves + 4 spines
WEKA	12 backends / ~1.09 PB	24 backends / ~2.17 PB	48 backends / ~4.34 PB
OOB	4 SN2201 access + 2 aggregation planned	6 SN2201 access + same aggregation	21 SN2201 access + 2 aggregation
Firewall	HA pair; initial capacity	Same security model	Same model; throughput expansion if required
Control plane	10-node service baseline	Reuse	Reuse / scale service roles as required
Observability	3-node minimum service pool	Expand toward 6	6-node planning baseline
Rafay	SaaS controller + cluster operator	Same control model	Same control model

13. Integrated Target-State BoM / BoQ Baseline

- The table below is the architecture-level target-state BoM/BoQ baseline.
- Architecture-driving quantities are fixed where the solution design is closed
- Supplier-specific part numbers, subscriptions and media remain explicitly conditional where vendor/customer confirmation or physical layout is still required.

Domain	Component	Qty	Specification / note
Compute	HGX B300 systems	128	8 x B300 GPUs; 8 x CX-8; target 1 x BF3 B3240 dual 400G
Compute	B300 GPUs	1,024	Included in HGX systems
E-W	Q3400-RA leaf	16	Quantum-X800 XDR 800G
E-W	Q3400-RA spine	8	Quantum-X800 XDR 800G
E-W	Active server-to-leaf links	1,024	800G XDR
E-W	Leaf-to-spine links	1,152	800G XDR; full 144-node fabric envelope
N-S/storage	SN5610 leaf	8	64 x 800G Spectrum-X
N-S/storage	SN5610 spine	4	64 x 800G Spectrum-X
N-S/server	BF3 B3240	128	Target 1/node; 2 x 400GbE. Supermicro supplied SKU requires confirmation.
N-S/server	B300-to-leaf links	256	400GbE
Storage	WEKA backends	48	Planning baseline: 10 x 15.36 TB NVMe/backend
Storage	WEKA 400G links	96	2/backend across independent leaves; exact NIC SKU to WEKA verification
OOB	SN2201 access	21	18 compute + 3 infrastructure
OOB	100G aggregation switches	2	Exact model TBD
Security	HA NGFW appliances	2	Vendor/model/throughput after customer edge-bandwidth confirmation
Control	x86 control/service nodes	10	2 BCM + 3 K8s admin + 3 K8s user/platform + 2 Slurm
Observability	x86 monitoring/logging nodes	6	Planning baseline; retention sizing may refine
Software	Rafay	1 service subscription	Node/cluster licensing to vendor quote
Software	WEKA	~4.34 PB planning usable	Exact licensed metric and support tier to WEKA quote
Software	Kubernetes / Slurm / NVIDIA stack	As deployed	Support/subscription SKUs according to enterprise support model

Software	Observability platform	1 platform	Prometheus/Grafana/DCGM plus centralized logging and SIEM integration; commercial/licensing model TBD

14. Optics and Cabling Engineering Baseline

- Optics counts distinguish logical links, switch-port equivalents and physical transceiver positions.
- These values are an engineering baseline rather than an installation schedule: exact optical/DAC/AOC choices, cable lengths and final part numbers depend on rack geometry and the qualified NVIDIA/WEKA media matrix.

Fabric	Connection	Logical links	Working transceiver basis
Quantum-X800	B300 CX-8 -> Q3400 leaf	1,024 x 800G	1,024 server-side modules + 512 Q3400 twin-port modules if optical
Quantum-X800	Q3400 leaf -> spine	1,152 x 800G	576 leaf-side + 576 spine-side twin-port modules if optical
Spectrum-X	B3240 -> SN5610 leaf	256 x 400G	256 node-side 400G + 128 switch-side 2x400G breakout modules
Spectrum-X	WEKA 400G NIC -> SN5610 leaf	96 x 400G	96 endpoint-side 400G positions + 48 switch-side 2x400G breakout positions; exact NIC/transceiver SKU to WEKA/supplier verification
Spectrum-X	SN5610 leaf -> spine	256 x 800G	512 switch-end 800G transceiver positions; exact qualified cable assembly basis to supplier confirmation
OOB	SN2201 -> OOB aggregation	42 x 100G	84 x 100G transceiver endpoints if pluggable optics are used

15. Facilities Engineering Scope

- Detailed data-centre facilities engineering is outside this Solution Architecture.
- Final rack elevations, electrical distribution, breakers, intelligent rack PDU selection, liquid-cooling/CDU design, hydraulic calculations, heat rejection, structural loading, UPS/generator/utility sizing and site commissioning are to be completed by qualified facilities engineers in coordination with Supermicro, NVIDIA NVIS and the selected data-centre operator.
- This proposal hands off the IT equipment quantities, OEM power/thermal envelopes, liquid-cooling dependencies, A/B power expectation, management interfaces and indicative equipment populations required to support that downstream facilities workstream.

16. Remaining Customer / Vendor Confirmations

Priority	Confirmation / decision	Required outcome and proposal treatment

P0	Supermicro confirm exact BlueField-3 SKU in candidate B300 stock and availability of the target B3240 dual-400G configuration	Confirm whether candidate stock meets the target 2 x 400GbE envelope. Any lower-bandwidth DPU remains a documented compliance deviation.
P0	ACME: confirm external/customer/WAN bandwidth and security-service requirements	Select final firewall throughput class and populate the external/core uplink envelope.
P0	WEKA: verify the 12/24/48 backend profile, SSD SKU, protection geometry and aggregate 250/100 GB/s target	Promote the storage design from bounded architecture sizing to vendor-verified procurement sizing.
P1	ACME: confirm single-tenant vs multi-tenant operating model	Finalize VRF/VLAN segmentation, IAM, quota and workload-isolation policies.
P1	ACME: confirm upstream Kubernetes baseline or require OpenShift	Finalize subscription, lifecycle and platform-operations model.
P1	ACME: confirm log, metric and security-event retention periods	Finalize observability/logging capacity and licensing.
P2	Physical design team / suppliers: confirm rack geometry, cable lengths and qualified optics/media	Freeze final cable and optics SKUs as part of the detailed physical design / installation package.
P2	ACME: confirm monitoring retention, enterprise ITSM/NOC integration and target SIEM/SOC endpoints	Finalize observability/logging capacity, retention policy, integrations and licensing.

17. Architecture Assurance and Engineering Evidence

The architecture has been independently regenerated and reconciled through the governed architecture workflow for all three population states. Architecture, physical-realization, security, BoM/BoQ and document outputs reconcile to the same revision-bound inputs. Detailed network evidence is maintained separately from the concise customer diagrams so that engineering teams can inspect exact physical connectivity without overloading this proposal.

Assurance domain	Phase 1 - 8 B300	Phase 2 - 16 B300	Phase 3 - 128 B300
Architecture population	Validated	Validated	Validated
HGX B300 compute quantities	Validated	Validated	Validated
Quantum-X800 topology	Validated	Validated	Validated
Spectrum-X / storage topology	Validated	Validated	Validated
WEKA capacity projection	Anchor	Bounded candidate	Bounded candidate
Physical device/port/link realization	Validated	Validated	Validated
Security zones and governed flows	Validated	Validated	Validated
Cumulative / incremental BoM-BoQ	Validated	Validated	Validated

Architecture documents	Validated	Validated	Validated
Detailed vector engineering views	Validated	Validated	Validated
Vendor procurement authority	Not granted	Not granted	Not granted
Facilities / installation authority	Not granted	Not granted	Not granted

Note Supporting engineering evidence includes deterministic fabric views, per-switch connectivity views, link-trace sheets and machine-readable ledgers. At final scale the evidence reconciles 2,176/2,176 Quantum-X800 links and 608/608 Spectrum-X links with exact endpoint and port identities. These artifacts support build-out review and detailed engineering but do not grant procurement or installation authority.

18. Proposal Issue Position

- The solution architecture is complete for customer review.
- The remaining P0/P1/P2 items in Section 16 are explicit confirmation gates rather than unresolved architectural design gaps.
- Unless Supermicro or WEKA identifies a material incompatibility, those confirmations refine component compliance, procurement sizing, licensing and edge capacity without changing the fundamental compute, east-west, north-south/storage, OOB, security or software-control architecture.

Recommendation:

- **Proceed with detailed vendor confirmation and commercial quotation against this architecture baseline.**
- **Freeze the final BlueField SKU, WEKA procurement profile and external firewall/uplink sizing before purchase order issue**
- **Then hand the approved equipment set to detailed facilities and installation engineering.**

19. Baseline References

- NVIDIA Enterprise Reference Architecture - HGX B300 servers and Spectrum-X networking (2026).
- NVIDIA DGX B300 / Quantum-X800 scalable infrastructure reference architecture (2026).
- WEKA Qualified Program Nitro Reference Design for NVIDIA HGX H200/B200/B300 servers (2026).
- Rafay Kubernetes Operations Platform architecture and network-whitelisting documentation.
- ACME/XingYun source architecture and BoQ gap analysis supplied for this engagement.
- Neon AI Factory Architect governed architecture, WEKA knowledge and physical-realization evidence used to reconcile architecture quantities and engineering traceability.